

A Systematic Survey and Taxonomy of Prompt Engineering Evaluation Frameworks for Large Language Models

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Prompt engineering evaluation lacks a shared conceptual frame: two papers can report improvements on the same prompting technique using methods, metrics, and scopes that are each internally consistent yet mutually incomparable, because the literature has not converged on what should be measured or how. This survey provides a measurement-theoretic taxonomy of prompt-engineering evaluation, tracing the incomparability to two long-standing tensions in measurement theory: construct validity vs. measurement reliability, and reference-anchored vs. judgment-anchored evaluation.

We propose a seven-dimensional taxonomy of the evaluation choices a study makes: how outputs are scored, on what task, with which metric, how automated the pipeline is, over how many models and datasets, whether rephrasing the prompt was tested, and whether a third party could reproduce the result. We apply it to a corpus of 151 papers assembled by a PRISMA 2020 systematic review. The coding falls into three recurring configurations, which we name Benchmark-Automation, Judge-Mediated, and Expert-Anchored, and which are largely a compact summary of how a study scores its outputs. We report two negative results about them: the dimensions the archetype rule does not read carry no independent trace of the partition, and the archetypes are not associated with held-out robustness or reproducibility practice once a structural dependency is removed. The same coding reveals three recurring failure modes, each a configuration that does not fully support the claim the paper makes with it; 45.0% of the corpus exhibits at least one. We ask readers to use the failure-mode catalog and the companion seven-question design checklist, which are the parts that survive this scrutiny intact.

Benchmark-Automation papers (66.2%) live in the high-reliability, narrow-validity region: fine for ground-truth tasks, under-specified for open-ended generation. Judge-Mediated work (17.2%) addresses the construct gap, but the corpus has not converged on a shared judge-calibration protocol. Robustness and contamination-resistant evaluation remain minority practices in the corpus. Reporting a result's sensitivity to rephrasing, and the exact version of any LLM judge, should become default requirements rather than optional extras.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; *Language resources*; • **Information systems** → *Evaluation of retrieval results*; Test collections; • **General and reference** → *Surveys and overviews*.

Additional Key Words and Phrases: prompt engineering, large language models, evaluation frameworks, taxonomy, systematic review, LLM evaluation, benchmark evaluation, LLM-as-judge, evaluation design, evaluation checklist, NLP benchmarks, automation level, task scope, metric selection, evaluation reproducibility

1 Introduction

Large language models (LLMs) underpin production software developed over the past four years. Prompt engineering (PE), crafting natural-language inputs to elicit reliable outputs without modifying weights [70, 104], is the interface most teams use to control them. Chain-of-thought, retrieval-augmented generation, automated prompt optimization, and dozens of related techniques are deployed from arithmetic reasoning to clinical decision support [61, 99, 135]. Figure 1 previews

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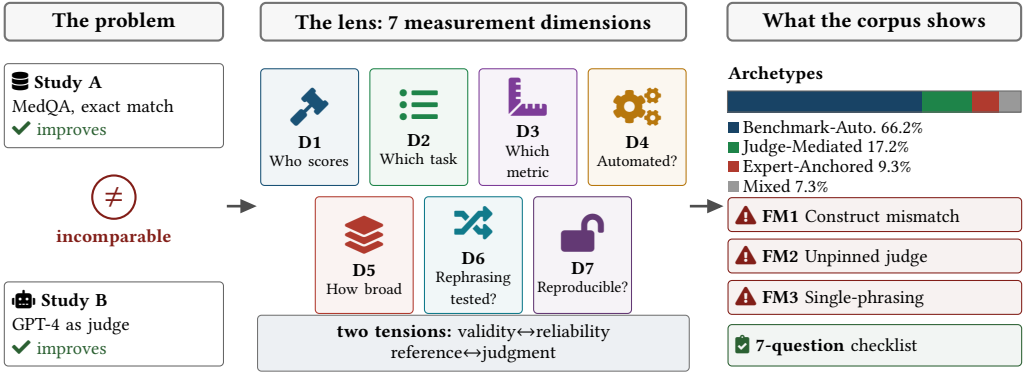


Fig. 1. Two studies of the same prompting technique can each report an improvement while remaining incomparable, because they differ in what scores the output, on what task, and how. The seven-dimensional lens developed in this survey makes those differences explicit and shows, empirically, that most papers in our corpus fall into one of three recurring configurations.

why: two evaluations of the same technique can both report improvement and still be incomparable, which the seven-dimensional lens developed below explains.

A concrete scenario shows why this matters. A chain-of-thought variant for medical differential diagnosis is evaluated by Study A on MedQA with exact-match accuracy, and by Study B with GPT-4 acting as a clinical scoring judge on a different question set. Both report improvements, but a clinician cannot combine the results: A’s exact-match score does not capture clinical reasoning or safety, and B’s judge score is uninterpretable without calibration. Neither study fails internally. Both are limited by an evaluation ecosystem without a shared vocabulary for what each measured and when results transfer [23, 46]. This paper builds that vocabulary.

Two questions decide whether Study A and Study B can be compared at all: does the score measure what the paper claims (A’s exact-match number is cheap to reproduce but silent on clinical safety), and does re-running the evaluation return the same score (B’s judge tracks what a clinician cares about, but moves when the judge model updates)? These goals pull against each other: the easiest scores to reproduce are often furthest from what we want to know. A second split cuts across the first: a score is either fixed by a stored answer key, or produced by an evaluator, human or model, that reads the output and forms an opinion, which we call a *judge*. From these two pulls we derive seven questions we code every paper against: what scores the outputs, on what task, with which metric, how automated the pipeline is, how many models and datasets it covers, whether rephrasing the prompt was tested, and whether a third party could reproduce the result.

This paper reorganizes prompt engineering evaluation practice into a measurement-theoretic frame, borrowing the validity and reliability constructs that psychometrics developed for human testing [30, 78] and that recent work has begun applying to machine learning [54]. We add the empirical work needed to test whether the reorganization is well-posed.

Building this taxonomy and corpus was harder than listing categories. **Coding choices papers rarely state explicitly.** Most papers do not name their method, metric family, or automation level directly; coding one meant inferring these from scattered methods-section prose, tables, and released code. **A corpus that can bias its own distributions.** Five PRISMA expansion rounds, each chasing a different coverage gap, could each shift which evaluation styles were over- or under-represented before any paper was coded. **Validating a taxonomy without circularity.**

The archetypes we report are found in the same codes the taxonomy defines, so trusting them needs evidence beyond the assignment rule that produced them (Section 8.10.1).

The contribution is four deliverables, each stated as a practice a future evaluation paper could adopt, and each verifiable on its own:

- (1) *A way to record a study's evaluation choices in one comparable line.* A reproducible codebook that records the seven evaluation choices above, each as one option from a fixed set, with reproducibility recorded as three yes/no flags. Two coders applied it independently to a subsample of 38 papers and agreed at Cohen's κ of 0.76–0.92 (Section 7.5).
- (2) *A frame against which a future evaluation paper can state its own tradeoffs.* The two pulls just described, stated precisely and carried through the whole paper as the reason the seven dimensions are the ones we chose.
- (3) *A check to run before trusting a taxonomy's own clusters.* Papers do not spread evenly across the possible configurations. Almost all of them fall into three recurring bundles, which we name Benchmark-Automation, Judge-Mediated, and Expert-Anchored. We assign them with a rule that reads only how a study scores its outputs, and we report honestly what that buys: the bundles summarise scoring practice compactly, but the other dimensions carry no independent trace of them (Section 8.10.1).
- (4) *A checklist for catching an evaluation design that cannot support its own claim.* Three recurring evaluation-design mistakes, each one a specific combination of choices that leaves a paper unable to support the claim it makes. We name each mistake, and point at the corpus papers that make it (Section 8.11).

Three streams of prior work each supply part of what we need. Surveys of prompting techniques [73, 99, 104, 121] sort methods into labelled categories. We borrow that habit, but point it at how techniques are *evaluated* rather than at how prompts are written. Surveys of LLM evaluation [23, 44, 83, 84] organize benchmarks, agents, and LLM judges along their own axes. We keep the idea of dimensional axes but choose axes specific to prompt engineering, and we treat robustness and reproducibility as questions worth asking of every paper rather than as caveats mentioned at the end. Third, a recent line of work asks whether a benchmark score really measures the capability it claims to [6, 12, 144]. This is the construct-validity literature: it imports into machine learning a question psychometrics has asked of human tests for seventy years, namely whether a test measures the thing its designer says it does. We ask that question of evaluation setups rather than of benchmarks. Joining the three streams gives a codebook that classifies evaluation choices rather than prompts, three recurring configurations that summarise how a study scores its outputs, and failure modes tied to named papers. Each stream is insufficient on its own; Section 2 says what each is missing.

Research Questions

This survey is organized around four research questions:

- RQ1** (*Taxonomy.*) What dimensions, if any, suffice to classify the evaluation framework of a prompt-engineering study in a way that makes its measurement trade-offs visible? Addressed in Sections 4 and 4.1.
- RQ2** (*Distribution.*) What are the empirical distributions of evaluation choices across the 151-paper corpus, and which combinations co-occur often enough to constitute recurring *archetypes*? Addressed in Sections 8.1–8.9 and 8.10.
- RQ3** (*Failure modes.*) Where does the taxonomy expose specific points at which an evaluation configuration does not support the paper's contribution claim, and how prevalent are these in the corpus? Addressed in Section 8.11.

RQ4 (*Tensions and gaps.*) What interpretive tensions appear consistent with the observed configurations, and what gaps in reporting practice does the corpus show? Addressed in Sections 9.1–9.3 and the future-direction agenda of Section 9.6.

Paper Organization

Related work (Section 2) and background (Section 3) precede the taxonomy itself (Sections 4–6), the review methodology (Section 7), the corpus results (Section 8), and the discussion and conclusion (Sections 9–10). The Electronic Supplement holds the evaluation-design checklist (Appendix A) and the full 151-paper coding (Table C.1).

2 Related Work

2.1 Prompt Engineering Methods and Techniques

Interest in systematic prompt design grew once input phrasing was shown to shift model outputs [19]. Few-shot [19] and chain-of-thought [135] established the foundational paradigms; the technique catalog now includes zero-shot CoT [58], self-consistency [132], tree- [142] and graph-of-thought [14, 15], least-to-most [152], plan-and-solve [129], faithful CoT [75], many-shot ICL [3], AutoPrompt [108], FLAN [28], prompt-pattern catalogs [136], manual-prompt studies [40], RAG [61], and automated optimization (GEPA [4], PromptWizard [2]). Recent automated-prompt-optimization surveys [65, 96] and prompting-technique catalogs [99, 104] organize this territory but leave the evaluation question open: how is a working prompt actually verified?

2.2 Evaluation Frameworks for Large Language Models

LLM evaluation developed along its own trajectory: GLUE/SuperGLUE [127, 128] for NLU, BIG-Bench and BIG-Bench Hard [111, 114] for reasoning, MMLU [49] for multitask knowledge, and MT-Bench/AlpacaEval [34, 150] for open-ended instruction following with LLM-based judges. Reference-free and model-based scoring [93, 139, 147, 151], in-context retrieval [95], and prompt-format studies [102] expand the design space. Ni et al. [84] reviewed 283 benchmarks and found recurring contamination, cultural bias, and memorization problems; Gu et al. [44] document LLM-judge bias at scale, and Feng et al. [39] find even top judges inconsistent on a quarter of difficult pairs. A benchmark-quality checklist [98] and lm-eval-harness setup-sensitivity reports [16] document the same non-reproducibility this survey treats as D7, in a landscape of plentiful tools with no shared structure for comparing across them.

2.3 Surveys and Taxonomies in Prompt Engineering and LLM Evaluation

Survey literature treats prompting methods and evaluation practice as separate topics with no systematic bridge. On the PE methods side, Schulhoff et al. [104] catalog 58 prompting techniques (meta-analysis, no evaluation framework); Sahoo et al. [99] and Vatsal & Dubey [121] organize methods by application domain and NLP task type respectively; Liu et al. [73] build a four-aspect taxonomy of techniques (profile/instruction, knowledge, reasoning/planning, output-bias reliability) that mirrors our title structure but classifies prompts not their evaluations; Fagbohun et al. [38] provide a practitioner guide; Sasaki et al. [103] and Tian et al. [117] cover SE-specific patterns and code prompt defects; He et al. [48] situate PE within “context engineering”; Liu et al. [71] survey prompting frameworks as toolkits; Plaat et al. [89] survey multi-step reasoning; and Dong et al. [33] give the foundational in-context-learning survey of techniques, whose scope excludes ICL evaluation methodology, our subject. Of the eleven CSUR-adjacent surveys we compare against in Table 1, none taxonomizes the *evaluation methodology* under which PE techniques are assessed:

the nearest among them cover prompting toolkits [71] and the prompting paradigm [70], the gap this work fills.

On the evaluation side, Chang et al. [23] organize LLM evaluation around what/where/how; Zhao et al. [149] and Hou et al. [50] cover explainability and LLM-for-SE evaluation respectively; McIntosh et al. [77] critique benchmark adequacy; Ni et al. [84] sort 283 benchmarks into a three-tier taxonomy; Mohammadi et al. [83] and Yehudai et al. [146] propose agent-evaluation taxonomies; Guan et al. [45] apply a PRISMA-inspired review to ~250 multi-turn-agent sources; Li et al. [62] survey LLM-as-Judge; and Lu et al. [74] propose a linguistic *PromptPrism* taxonomy of prompt structure.

Four works share a measurement-theoretic framing at other scopes: Bean et al. [12] review 445 LLM benchmarks for construct validity; Alaa et al. [6] focus on medical benchmarks; Ye et al. [144] survey LLM evaluation through the psychometrics lens; and Kearns [56] quantifies construct validity across the OpenLLM Leaderboard. Budagam et al. [20] score individual prompts, not study-level evaluation-framework configurations. Three works share our failure-mode framing at a different unit of analysis: Alampara et al. [7] apply measurement theory to materials-science ML (not PE); Qi et al.’s RIFT [92] taxonomizes *rubric* design failures against our PE *evaluation-configuration* failures; and Vinay [125] catalogs production LLM-*system* failures against our *evaluation-methodology* failures (Section 2.4).

Three surveys published in the same period are the closest neighbours of this one. Cantini and De Mauro [22] run a PRISMA-based review of structured prompting techniques, taxonomizing the techniques rather than their evaluation. Sheikhi et al. [107] propose a five-paradigm taxonomy of general LLM evaluation methodology; the overlap is real, but their scope is LLM evaluation at large, not the PE-specific choices coded here. Botter’s psychometric critique of LLM benchmarks [18] draws on the same theoretical toolkit at the general-benchmark level; this survey applies that toolkit to the evaluation choices of prompt-engineering studies.

2.4 Differentiation from Prior Surveys

Several recent surveys overlap in topic with this paper; the question is what the integration produces beyond reorganizing known evaluation practices. Table 1 positions this work against eleven closely related surveys on three axes: (1) *primary scope* (catalogs prompting methods, evaluation methods, or both), (2) *theoretical framing* (whether the survey commits to a conceptual structure beyond a flat enumeration), and (3) *coverage of the seven evaluation questions* previewed in Section 1. Each prior row supplies a piece we build on; no row covers all seven questions.

Three differences are central. First, no compared survey organizes PE *evaluation* around a measurement-theoretic frame. Bean et al. [12] and Ye et al. [144] apply construct-validity and psychometric framing respectively, but to LLM benchmarks broadly and LLM capabilities; Chang et al.’s what/where/how is descriptive, not tension-based. Second, no compared survey treats robustness and reproducibility / contamination resistance as first-class dimensions of evaluation design (Ni et al. on contamination and Gu et al. on judge bias treat them as sub-findings). Third, prior PE-focused surveys with similar titles (Schulhoff, Sahoo, Vatsal & Dubey, Liu et al. 2026) catalog *methods* not *evaluation frameworks*; the title-overlap with Liu et al. 2026 is purely lexical (their four aspects classify prompt designs; our seven classify the evaluation configurations under which prompts are tested). Budagam et al. [20] use “evaluation framework” for a single hierarchical prompting scheme, not a taxonomy of existing evaluation practice. This survey is positioned in the gap left by all three groups: a measurement-grounded taxonomy of PE *evaluation* that explicitly incorporates robustness and reproducibility.

Table 1. Differentiation from prior surveys. “Methods” refers to a catalog of prompting techniques; “eval” refers to a catalog of evaluation methods, benchmarks, or metrics. “Theoretical framing” means commitment to a conceptual structure (e.g. a tension, a paradigm, or a measurement-theoretic axis) rather than a flat taxonomy. ✓ = covered as a primary axis; ○ = mentioned but not classified along; – = not covered. The seven right-hand columns are the seven evaluation questions previewed in Section 1 and formalized as dimensions D1–D7 in Section 4. This paper is the only one combining a PE-technique catalog, an evaluation-method catalog, and a measurement-theoretic frame applied to PE. Closest title overlap: Liu et al. [73] (techniques, not evaluations); closest framing overlap: Bean et al. [12] and Ye et al. [144] (LLM benchmarks/capabilities, not PE).

Survey	Primary scope	Theoretical framing	Method	Task	Metric	Autom.	Scope	Robust.	Reprod.
Schulhoff et al. [104]	Methods	Flat catalog (58 techniques)	–	✓	–	–	–	○	–
Sahoo et al. [99]	Methods	Application domain	–	✓	–	–	○	–	–
Vatsal & Dubey [121]	Methods	NLP task type	–	✓	–	–	–	–	–
Liu et al. [73]	Methods	Profile / Knowledge / Reasoning / Reliability (output bias)	–	✓	–	–	–	–	–
Budagam et al. [20]	Eval (prompting)	Hierarchical 5-level prompting framework, cognitive principles	–	✓	–	–	–	–	–
Chang et al. [23]	Eval	What/where/how to evaluate	✓	✓	✓	○	○	○	–
Ni et al. [84]	Eval (benchmarks)	Three-tier taxonomy of 283 benchmarks	○	✓	○	–	–	–	○
Mohammadi et al. [83]	Eval (agents)	Two-dimensional	✓	✓	○	–	○	–	–
Gu et al. [44]	Eval (LLM-judge)	Bias / reliability of LLM judges	○	–	✓	–	–	○	○
Bean et al. [12]	Eval (benchmarks)	Construct validity (445 benchmarks, 29 reviewers)	○	✓	✓	–	–	○	○
Ye et al. [144]	Eval (LLMs)	Psychometric validation & enhancement	○	✓	✓	–	–	○	–
This work	Eval (PE)	Validity / reliability tensions, archetype clustering, failure modes	✓	✓	✓	✓	✓	✓	✓

3 Background

This section fixes the vocabulary the taxonomy is built from: what a prompt-engineering evaluation involves, the two measurement-theory tensions the dimensions are organized around, and the seven formal definitions the rest of the paper refers back to.

3.1 LLMs, Prompting, and the Design Space

LLMs are transformer-based [120] networks trained on web-scale text that handle broad task ranges without task-specific retraining [17, 19, 32]. Practitioners interact through prompts, and surface

phrasing produces nontrivial output differences [105, 154]. Models can even perform well under prompts that are semantically irrelevant or misleading [134], a construct-validity warning that a prompt’s manifest content and the capability it is credited with eliciting can decouple. Prompt engineering covers methods spanning manual to fully automated: zero-shot/few-shot [19], chain-of-thought [135], retrieval-augmented generation [61], and automated prompt optimization via textual gradient descent [91] or LLM-based search [154]. The cross-cutting question, “how do you know it worked?”, depends on the evaluation framework, and those frameworks are what the taxonomy in the next section classifies.

3.2 Evaluation Metric Families

Four metric families recur: **reference-based** (BLEU [88], ROUGE [69]) score by surface overlap, fast but weakly aligned with human judgment on open-ended tasks [42, 72]; **model-based** (BERTScore [148]) use learned semantic representations but still require a reference [23]; **LLM-as-Judge** (G-Eval [72]) delegate scoring to a second LLM, eliminating the gold-reference requirement at the cost of judge biases [44, 150]; and **task-specific** (pass@ k , exact match, RAGAS [37]) are purpose-built and rarely transfer. Section 4 organizes these along D3.

3.3 Validity and Reliability in Evaluation

Section 1 introduced the two pulls in plain words. Here we state them in the sense measurement theory gives them. *Reliability* is consistency under repetition. *Construct validity* is the degree to which a score is evidence about the property we actually care about [30, 78]. The two typically pull against each other: exact-match scoring against a fixed key is highly reliable but narrow on “answer quality”; expert-panel scoring of open-ended outputs pursues higher construct validity at the cost of speed and inter-rater agreement [23]. Twenty years of human evaluation in NLP shows the same criteria named and measured inconsistently across studies, motivating a fixed vocabulary before coding a corpus against it [51]. Psychometrics developed both constructs for human testing, and a recent line of work carries them into ML and NLP evaluation [54, 67, 113], including a measurement-theoretic formalization of NLG metric reliability and validity [137] and a four-level account of what measuring a generative system responsibly requires [126]. The seven dimensions in Section 4 are built on that translation.

3.4 Formal Definitions

These seven definitions are inputs to the coding, not results of it: we wrote them first, then coded the corpus against them. Definitions 2 and 3 restate construct validity and reliability as psychometrics has used them for decades [30, 78]. Definitions 4 and 5 name the two ends of the reference/judgment split, which the LLM-judge literature [44, 150] already works with informally. Definitions 6 and 7 are the ones we add: they take concerns that the prompt-sensitivity [82] and contamination [100] literatures raise in passing and make them properties a paper either has or lacks. Definition 1 introduces the framework object $\mathcal{F}$, and Definitions 2–7 each state a property of that same $\mathcal{F}$, which is why the symbol recurs throughout.

DEFINITION 1 (PE EVALUATION FRAMEWORK). *A prompt-engineering evaluation framework $\mathcal{F}$ is one complete set of answers to the seven questions above: what scores the outputs, on what task, with which metric, how automated the pipeline is, over how many models and datasets, what was reported about rephrasing, and what was reported about reproducibility. Any complete set of answers is a valid configuration; the corpus in Section 8 is analyzed as a distribution over such configurations.*

DEFINITION 2 (CONSTRUCT VALIDITY). *An evaluation configuration $\mathcal{F}$ has construct validity for a target property π to the degree that high scores constitute evidence that a model or technique achieves*

π in the intended deployment context. Construct validity is degree-valued: arithmetic benchmarks have high validity for arithmetic, lower validity for open-ended clinical reasoning [30, 54, 78]. GDP is the familiar case: it is measured to many decimal places and reported quarterly, and it remains a contested measure of national well-being. A number can be trustworthy and beside the point at the same time.

DEFINITION 3 (MEASUREMENT RELIABILITY). $\mathcal{F}$ is reliable if repeated applications to the same model-prompt pair produce scores with small variance under nominally identical conditions. Within-run (fixed software and model snapshot) and across-time (across API revisions, judge updates, benchmark releases) reliability are distinct; high within-run reliability does not imply across-time reliability [113].

Terminology. There is one notion of *judge* in this paper, and Definitions 4, 5, and 7 all use it: a judge is any evaluator, human, expert panel, or LLM, whose verdict is not fixed in advance by a stored answer key. When the judge is itself an LLM we say *LLM judge*. Recording which LLM judge was used is the D7 sub-property *JP* in Definition 7.

DEFINITION 4 (REFERENCE-ANCHORED EVALUATION). $\mathcal{F}$ is reference-anchored if scoring is determined by comparison to a stored gold answer (exact-match key, reference translation, passing test suite) without a separate judge. Reference-anchored configurations have high within-run reliability but fail when no single correct answer exists, the reference is contaminated by training data, or the reference under-determines acceptable outputs.

DEFINITION 5 (JUDGMENT-ANCHORED EVALUATION). $\mathcal{F}$ is judgment-anchored if scoring is delegated to an evaluator (human, expert panel, or LLM judge) whose output is not pre-determined by a stored reference. These configurations can pursue construct validity beyond what fixed references support but introduce evaluator biases and require calibration for cross-study interpretability [39, 44].

DEFINITION 6 (ROBUSTNESS REPORTING). A paraphrase is a reworded prompt that a competent reader would agree asks for the same thing: “What is 17+25?” and “Compute the sum of 17 and 25” are paraphrases of each other. Rewording acts on the input, so paraphrases exist even when the output is a single number. Let k be the number of paraphrases a paper evaluates its technique on; when the evaluation is judgment-anchored, k may count paraphrases of the technique prompt or of the judge’s scoring prompt, and we record which. Then $\mathcal{F}$ reports robustness at tier r , where $r = 0$ if robustness is not reported at all; $r = 1$ if fewer than five phrasings are used ($1 \leq k < 5$); $r = 2$ if results are reported for $k \geq 5$ phrasings, whether aggregated into the headline or in a separate robustness study; and $r = 3$ if the headline pipeline additionally uses the resulting spread to bound how much the reported number could move. The tiers are ordered and exhaust all k . We record these four tiers as the D6 subcategories Not-reported, Single-prompt, Multi-prompt aggregate, and Sensitivity-bounded (§4.8) [82, 90].

DEFINITION 7 (REPRODUCIBILITY / CONTAMINATION RESISTANCE). $\mathcal{F}$ carries three non-mutually-exclusive D7 sub-properties: artifact release (AR: prompts, seeds, scoring code recoverable), contamination checking (CC: explicit verification the test set was not seen during pretraining), and judge-version pinning (JP: any LLM-judge API version or checkpoint recorded). AR supports within-run reproducibility; across-time additionally requires JP for judgment-anchored and CC for static-benchmark configurations.

4 A Taxonomy of Prompt Engineering Evaluation Frameworks

This section defines the taxonomy. We first state the two tensions that organize it, then define the seven dimensions D1–D7 one at a time, each as the answer to one evaluation-design question.

4.1 Two Interpretive Tensions

Two tensions organize the seven dimensions: (i) the construct-validity vs. measurement-reliability distinction long established in psychometrics [30, 78] and recently re-applied to ML measurement [12, 54, 144], and (ii) the reference- vs. judgment-anchored distinction implicit in the LLM-judge survey literature [44, 62]. What we add is their application to the dimensional choices PE evaluation studies make: construct validity vs. measurement reliability (Definitions 2–3), and reference-anchored vs. judgment-anchored evaluation (Definitions 4–5). We adopt these two tensions; Section 8.10 shows the corpus falling into three configurations that sit where the tensions predict, and reports why that pattern is weaker support than it looks. Figure 2 positions the dimensions; Section 8 locates the 151 papers.

What the frame organizes. The two tensions make three corpus patterns interpretable. Evaluation method and metric type travel together, strongly enough to dominate how often dimension pairs travel together (Section 8.9), because the judgment locus fixes the metric family. Benchmark-automation dominates closed-form tasks, where automation costs little construct validity; LLM-judge adoption fragments because an uncalibrated proprietary judge breaks comparability across papers.

4.2 From Five Dimensions to Seven

To be plain about provenance: we did not invent D1–D5, and no single prior paper defines all five. Each of the five recurs, under varying names, across existing surveys of prompting and LLM evaluation [23, 84, 99, 104]; different surveys carry different subsets. Our step was to consolidate them into one codebook with a fixed option set per dimension. D6 (robustness) and D7 (reproducibility) are the additions we do claim, and the claim is narrow. The prompt-sensitivity [36, 82, 97, 101, 105, 123, 156] and benchmark-contamination [10, 39, 77, 84] literatures both treat these as serious concerns, and the surveys in Table 1 mention them. What is new here is carrying both as dimensions on which every paper takes a recorded position, rather than as caveats raised in discussion.

Three further candidates are deliberately not dimensions: *calibration* is a property of D3 rather than its own axis; *uncertainty quantification* is subsumed by D6; and *deployment realism* is spread across D1, D2, and D5, though a later revision should promote it if agentic evaluation keeps growing.

Per Definition 1, an evaluation framework is fully specified by a choice along each of D1–D7. The dimensions are conceptually distinct but not statistically independent (Section 8.9); Table 2 summarizes them with subcategories, axis projection, and a representative example.

4.3 Dimension 1: Evaluation Method

D1 captures how a prompting technique is assessed. It projects primarily onto the judgment-locus axis: reference-anchored at the Benchmark and Ablation end, judgment-anchored at the Human, LLM-judge, and User-study end. Five subcategories appear in the corpus, each indexed to a primary strength and a primary weakness: *Benchmark* (reproducible, saturation-prone [49, 84]); *Human eval* (credible for open-ended tasks, expensive and slow to replicate [23]); *LLM-as-Judge* (scales to open-ended outputs, imports judge bias [34, 44, 150]); *Ablation* (isolates component contribution, not for cross-technique comparison); *User study* (deployment-context evidence, rare in the NLP literature). LLM self-evaluation/self-refinement methods (Constitutional AI [9], self-refine [76]) do not map cleanly onto any single subcategory because evaluator and evaluated system are the same model; we code such papers by their dominant external evidence (typically Benchmark or LLM-judge) and note self-evaluation as the studied mechanism.

4.4 Dimension 2: Task Scope

D2 captures the type of task the evaluation targets. It acts as a *moderator* on the validity-reliability tradeoff: the same metric yields different validity properties depending on whether the task admits a single gold answer or a distribution of acceptable answers. The eight primary categories are: *Reasoning* (arithmetic, symbolic, commonsense; anchor domain for chain-of-thought [135]); *NL generation* (summarization, dialogue, open-ended writing); *Code* (pass@ k against test suites [64]); *QA* (exact match or F1); *Clinical/domain* (medical QA, legal reasoning [109]); *Multilingual* [121]; *Agentic/tool-use* (trajectory success [141]); *Multimodal* (vision-, audio-grounded [41]). Papers spanning multiple categories are coded *Multi-task* as a cross-category label; *Agentic* is a smaller active population (4 papers) added in expansion rounds 2–3; *Multimodal* is defined for framework completeness but carries no active corpus instances.

4.5 Dimension 3: Metric Type

D3 captures what is being measured and by what means. It is the most direct projection of the reference-anchored / judgment-anchored axis: the four families introduced in §3, comprising reference-based (BLEU [88], ROUGE [69]), model-based (BERTScore), LLM-as-Judge (G-Eval [72]), and task-specific (pass@ k , exact match, RAGAS [37]), partition along that axis. Calibration evidence (validation of the metric against a target judgment on the specific task) is a D3 sub-property that becomes methodologically central for judgment-anchored metrics [44, 72].

4.6 Dimension 4: Automation Level

D4 captures how much human involvement the evaluation requires. It is a *sub-axis of reliability*: fully automated pipelines maximize across-run consistency at the cost of any construct validity human judgment would have added, while fully manual pipelines invert that tradeoff. We retain D4 separately because the corpus codes it independently of D1. The four subcategories are *Fully manual*, *Human-in-loop* (selective routing of borderline cases to reviewers [86]), *Hybrid* (automated bulk scoring with uniform human validation on a held-out sample), and *Fully automated*; human-in-loop and hybrid are not interchangeable, since the former applies judgment selectively on hard cases while the latter applies it uniformly.

4.7 Dimension 5: Evaluation Scope

D5 captures what is being compared. It measures how far a paper’s conclusions can extend beyond the specific setup that produced them. It sits on a generalizability axis only partly aligned with the validity-reliability frame: broader scope tends to come with more robustness and reproducibility reporting, though the association is moderate rather than tight (Section 8.9). The five subcategories run from narrow to broad: *Single-model, single-dataset* (one technique, one model, one benchmark; cleanest internally, narrowest claim); *Cross-model* (multiple model families, testing architecture-independence); *Cross-dataset* (task type fixed, benchmark varied, testing within-family robustness); *Cross-domain* (qualitatively different task types, the most demanding scope and the least common in individual papers, though survey works such as Sahoo et al. [99] cover it by design); and *Cross-lingual* (performance across languages [121]). Without explicit scope coding, the limits of single-model, single-dataset claims (which underpin a substantial share of PE-literature results) stay invisible. Scope also interacts with benchmark saturation: as benchmarks approach ceiling [49, 84], between-model spread shrinks and cross-model (D5) conclusions weaken, so D5 claims on saturated benchmarks warrant the D6/D7 checks.

Table 2. Seven-Dimensional Taxonomy of Prompt Engineering Evaluation Frameworks. Each dimension projects primarily onto one of the two underlying tensions introduced in Section 4.1: construct validity vs. measurement reliability (V/R), and reference-anchored vs. judgment-anchored evaluation (R/J). D5 projects onto a third generalizability axis (G) that is partially orthogonal to V/R: D5 is moderately associated with D6 and D7, and more weakly with D1–D4 (Table 12). D2 (task scope) is a moderator of V/R rather than an axial projection. Any combination of subcategories across dimensions is a valid evaluation framework configuration; co-occurrence patterns observed in practice are reported in Section 8.9. The seven dimensions index distinct design choices that prior surveys collapse into a single axis or omit; whether this finer resolution improves discrimination over a smaller axis set is examined in Section 8.10, which finds the archetype partition tracks D1 alone.

Dimension	Subcategories	Axis	What it captures	Representative example
D1 Evaluation method	Benchmark, Human eval, LLM-as-Judge, Ablation, User study	R/J	The procedure used to assess prompting technique performance	MT-Bench: LLM-as-Judge on open-ended tasks [150]
D2 Task scope	Reasoning, NLG, Code, QA, Clinical, Multilingual, Agentic, Multimodal, Multi-task	mod	The task domain that moderates the validity/reliability tradeoff	GSM8K: arithmetic reasoning [135]
D3 Metric type	Reference-based, Model-based, LLM-judge, Task-specific	R/J	The measurement approach used to score outputs (and its calibration)	G-Eval: GPT-4 rubric scoring [72]
D4 Automation level	Manual, Human-in-loop, Hybrid, Fully automated	V/R	Sub-axis of reliability: degree of human involvement	APO: fully automated scoring [91]
D5 Evaluation scope	Single-model, Cross-model, Cross-dataset, Cross-domain, Cross-lingual	G	The breadth of comparison across models, datasets, and domains	Sahoo et al.: cross-domain survey [99]
D6 Robustness	Not reported, Single-prompt, Multi-prompt aggregate, Sensitivity-bounded	V/R	Stability of reported performance under semantically equivalent rephrasings	PromptSET: paraphrase-based sensitivity benchmark [97]
D7 Reproducibility & contamination resistance	Artifact-released, Contamination-checked, Judge-version-pinned, Not-reported	V/R	Stability of results under judge drift, API version change, and benchmark contamination	WildBench: contamination-resistant dynamic benchmark [68]

4.8 Dimension 6: Robustness and Prompt Sensitivity

D6 captures whether the evaluation establishes that reported performance is stable under semantically equivalent rephrasings (Definition 6). Robustness projects onto the validity axis: a technique that degrades sharply under paraphrase is not, in a methodologically meaningful sense, the same technique the paper claims to evaluate. Empirical evidence for this concern is substantial [13, 97, 101, 105]; formatting changes alone can shift leaderboard rank by up to eight positions [8], and LLM-judge coherence flip rates exceed 60% on smaller judge models, consistent with independently reported judge self-inconsistency [47] and positional/familiarity bias [112, 130]. The four subcategories are the tiers of Definition 6 (Not-reported, Single-prompt, Multi-prompt aggregate, Sensitivity-bounded), ordered by reporting depth.

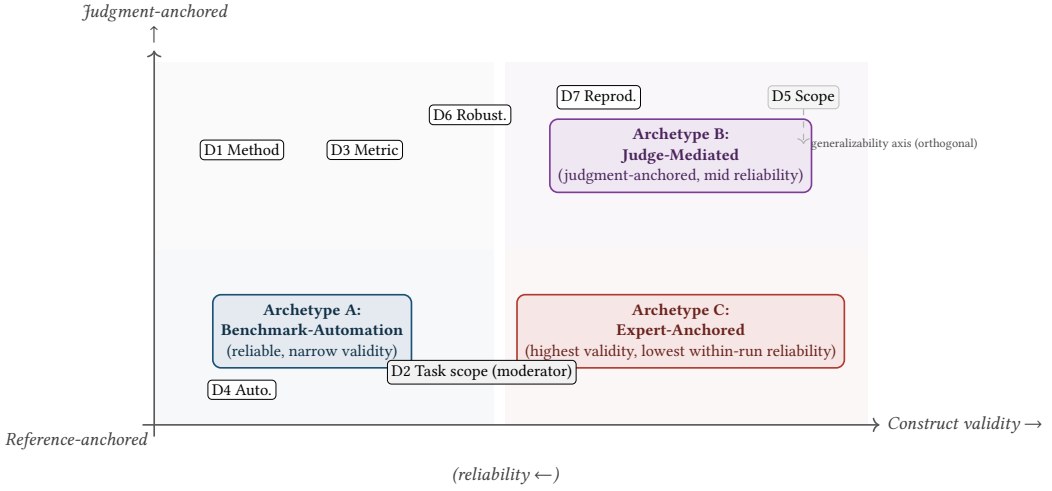


Fig. 2. Seven-dimensional taxonomy of prompt engineering evaluation frameworks positioned on the two underlying tensions introduced in Section 4.1. The horizontal axis is the validity / reliability tradeoff (left = high reliability, narrow validity; right = high construct validity, lower within-run reliability). The vertical axis is the judgment-locus tradeoff (bottom = reference-anchored; top = judgment-anchored). Read a dimension chip by its position, not by any connector: each chip sits in the region of the space that the dimension’s most common subcategory occupies. D5 projects onto a third generalizability axis (shown as a perpendicular annotation) and is not placed on the validity-reliability plane. The three shaded regions correspond to the recurring evaluation *archetypes* identified in Section 8.10; the empirical distribution of papers across them is reported in Table 7. The corpus clusters in the high-reliability, low-construct-validity quadrant (Benchmark-Automation archetype), suggesting benchmark-automation emphasizes measurement convenience over construct coverage.

4.9 Dimension 7: Reproducibility and Contamination Resistance

D7 captures whether a third party can reproduce the reported results and whether those results stay stable under infrastructure drift (Definition 7). It captures aspects of reliability that D4 does not: a pipeline can be within-run reliable (same call → same score today) and across-time fragile (judge model silently revised between API versions [39]; test items leaked into pretraining [84, 94]). The three non-mutually-exclusive *AR/CC/JP* sub-properties are coded as Definition 7 states them; *CC* additionally covers detection by guided-instruction contamination probes [43] and by held-out construction [68]. Papers reporting none of these properties are coded *Not-reported* (NR).

5 Applying the Taxonomy: Worked Examples and Design Guidance

Table 2 lists the valid options for every dimension: its “Subcategories” column is the exhaustive set a coder chooses from, mutually exclusive except at D7, and Sections 4.3–4.9 define each option in prose. Figure 2 places the dimensions on the two tensions; the rest of this section classifies two corpus papers (§5.1), walks a new study to surface implicit decisions (§5.2), and distills the guidance into a reference table (§5.3).

5.1 Classifying Existing Papers

Two widely cited corpus papers illustrate the classification, each as a seven-tuple a citing paper could embed in its own methodology section. **Wei et al., Chain-of-Thought [135]**: D1=Benchmark, D2=Reasoning, D3=Task-specific, D4=Automated, D5=Cross-model (PaLM 540B, GPT-3, LaMDA),

Table 3. Seven-Question Evaluation Checklist: one question per dimension. Each row names the tradeoff that dimension forces. Full preregistration checklist in Electronic Supplement (Appendix A). *Citation guidance:* work adopting this checklist as a preregistration or reporting template can cite this table (and Appendix A) as the source. A “no” answer is not a disqualifier but benefits from justification; without it, the dimension defaults to implicit benchmark choice rather than deliberate design.

Evaluation Design Question	Taxonomy Dimension + Key Trade-off
How will model outputs be scored?	D1 - Evaluation Method: precision of human judgment vs. scalability of automated pipelines (judgment-anchored vs. reference-anchored)
What type of task is being evaluated?	D2 - Task Scope: task-fidelity of a narrow domain vs. generalizability of a multi-task setting (moderates the validity-reliability tradeoff)
Which measurement operationalizes quality?	D3 - Metric Type: automation and reliability of reference-based metrics vs. construct validity of model-based or judge-based metrics
How much human effort is available?	D4 - Automation Level: scalability and within-run consistency of full automation vs. bias-control of human involvement (sub-axis of reliability)
Over how many models and datasets?	D5 - Evaluation Scope: narrow replicability of single-model evidence vs. generalizability claims that require cross-model, cross-domain, or cross-lingual coverage
Is performance stable under prompt rephrasing?	D6 - Robustness: cost of paraphrase-bounded evaluation vs. risk that reported gains do not survive semantically equivalent rewordings
Will third parties be able to reproduce the result?	D7 - Reproducibility: cost of pinning judge versions, releasing artifacts, and checking contamination vs. exposure to silent infrastructure drift and benchmark leakage

D6=Single-prompt, D7=Artifact-released. **Zheng et al., MT-Bench [150]**: D1=LLM-as-Judge, D2=Multi-task, D3=LLM-judge, D4=Automated, D5=Cross-model (28 models), D6=Multi-prompt aggregate (judge’s own setup), D7=Artifact-released, Judge-version-pinned. The codings flag a warning sign Section 8.11 returns to: Wei et al. make a cross-model claim from a single prompt phrasing, the pattern later called FM3; Zheng et al.’s coding does not.

5.2 Designing a New Evaluation

Consider a new retrieval-augmented generation (RAG) technique for clinical question answering. The taxonomy forces seven explicit design decisions *before* data collection: scarce gold-answer sets push *D1* toward human or hybrid evaluation over benchmark-only; *D3* rules out reference metrics (BLEU, ROUGE) without a gold reference in favour of model-based factual precision [80] or a clinician-calibrated LLM-judge; *D4* favours human-in-loop over full automation, since clinicians would catch systematic errors automation misses; *D6* needs sensitivity-bounded reporting over paraphrased clinical queries, since single-prompt is a generalizability risk in this domain; and *D7* favours judge-version-pinned and contamination-checked over not-reported, since medical benchmarks are small and easily contaminated. A fuller walkthrough, including D2 and D5, is in Appendix F of the Electronic Supplement.

5.3 Evaluation Design Quick Reference

6 Evaluation Tools and Resources

Twelve open evaluation frameworks map onto the seven-dimensional taxonomy. Benchmark-Automation tools (lm-eval-harness, Stanford HELM, OpenCompass, MMLU-Pro) maximize within-run reliability but cover narrow construct ranges. Judge-Mediated tools (MT-Bench, AlpacaEval 2.0, Prometheus, WildBench) broaden coverage at the cost of judge-side variance. Robustness frameworks (PromptBench, PromptSET, PromptEval, JudgeSense) explicitly quantify prompt-sensitivity but are not yet integrated into the primary evaluation tools. No single tool covers all seven dimensions, and the same split between benchmark-based, judge-based, and robustness-focused tooling recurs at the tool level that we later find among papers (Section 8.10). A full mapping (tool $\rightarrow$ supported D-dims, archetype, primary limitation) is in Appendix D of the Electronic Supplement.

7 Methodology: Systematic Literature Review

This section explains how we built the 151-paper corpus, and why that matters: every distributional claim in this paper is only as good as the sample it is computed on, so the sample has to be assembled by a rule someone else could re-run.

We follow PRISMA 2020 [87], the standard reporting checklist for systematic reviews, documenting how many papers were dropped at each stage and why. We searched six databases with a fixed query (§7.1), screened the results against explicit inclusion criteria with two reviewers agreeing at $\kappa = 0.81$ (§7.2), and coded each surviving paper along the seven dimensions, two coders agreeing at $\kappa = 0.76\text{--}0.92$ (§7.5).

The corpus was built in five rounds, and this shapes how every later count should be read: one broad primary search, plus four narrower rounds added to repair coverage gaps we noticed while coding. Those later rounds did not sweep for prevalence, so the corpus is partly purposive by construction, and we name the bias each round introduces in Section 9.5.

7.1 Search Strategy

We searched six databases: arXiv (cs.CL, cs.AI), ACM Digital Library, IEEE Xplore, Semantic Scholar, Scopus, and Web of Science [57]. The primary search ran in 2025-Q1–Q2; four expansion rounds followed in 2025-Q3 to 2026-Q2 (Table 4) with no lower year cutoff (nearly all retained papers postdate 2020).

The base query (adapted per database, full strings in the repository at <https://github.com/rohithreddybc/PromptEvalTaxonomy>):

("prompt engineering" OR "prompting technique" OR "prompt optimization" OR "chain-of-thought" OR "few-shot prompting" OR "zero-shot prompting" OR "retrieval-augmented generation") AND ("evaluation" OR "benchmark" OR "assessment" OR "metric" OR "framework" OR "human evaluation") AND ("large language model" OR "LLM" OR "GPT" OR "language model").

Supplementary searches targeted specific subtopics that the primary string was likely to undercount: LLM-as-Judge evaluation methodology, evaluation of automated prompt optimization systems, and cross-domain prompting benchmarks. We also conducted backward and forward citation searches on the ten most-cited papers in the initial results. Together, these supplementary searches recovered 22 papers that did not surface through the main database queries.

7.2 Inclusion and Exclusion Criteria

Included:

- Papers that propose, evaluate, or compare one or more prompt engineering techniques using a stated evaluation framework

- Papers that propose new evaluation frameworks, benchmarks, or metrics relevant to assessing prompted LLM outputs
- Survey and taxonomy papers covering prompt engineering or LLM evaluation
- Conference papers, journal articles, and arXiv preprints with at least one of: peer review, 10+ citations, or publication at a recognized NLP/AI venue (ACL, EMNLP, NeurIPS, ICLR, IEEE, ACM)

Excluded:

- Papers that use prompting as a method but do not report or discuss evaluation of the prompting approach itself
- Papers with no accessible full text
- Non-English papers

Re-admittance criterion (rounds 2–3). The initial round excluded papers whose *topic* was prompt injection, jailbreaking, or narrow clinical/legal application; rounds 2–3 re-admitted them when their *evaluation methodology* is what the taxonomy classifies (e.g., prompt-injection benchmarks and MIRAGE/MedRAG [138] contribute evaluation infrastructure). Re-admitted papers are flagged in Appendix C.

The use-as-tool exclusion is practically consequential: many papers use prompting incidentally without evaluating it; these do not contribute to understanding evaluation frameworks. The corpus spans five categories (PE techniques, benchmarks/datasets, LLM-as-Judge / metric-quality, PE/LLM-evaluation surveys, and sensitivity/format/code-prompting studies) unified by participation in PE evaluation infrastructure; recent CSUR systematic reviews use the same mixed-category convention [23, 35]. ArXiv preprints are retained only under one of three quality conditions (10+ citations, recognized venue, or citation-trace from an included peer-reviewed paper); the venue-criterion step excluded 73 records, and the two 2026 preprints in the corpus are flagged as preliminary. The 151-paper size is comparable to CSUR-tier systematic surveys (142 in [35], ~160 in [99], ~220 in [23]) and narrower than omnibus surveys because dimensional coding requires per-paper full-text review and dual coding. A precise peer-reviewed vs. arXiv split is not reported because many corpus papers were posted to arXiv first and later accepted at venues (NeurIPS, ACL, EMNLP, ICLR, TACL) while retaining the arXiv identifier; per-paper venue information is in the accompanying bibliography.

7.3 Screening Process

The primary-round database queries returned 2,847 unique records after deduplication. Two reviewers independently screened all titles and abstracts; disagreements were resolved through discussion (percent agreement 94.1%, $\kappa = 0.81$ over a 200-record dual-screened sample), with unresolved cases referred to a third reviewer (PRISMA 2020 Item 8 [87]). Title-abstract screening excluded 2,435 records; of the 412 advancing to full text, 283 were excluded (181 used prompting as a tool without evaluating it, 64 domain-specific applied, 24 inaccessible, 14 non-English). The remaining 129 papers passed full-text screening; the venue criterion excluded a further 73 records (two cross-round duplicates subsequently removed), and 22 papers recovered through citation searching brought the initial corpus to **78 papers**. Four supplementary expansion rounds (Table 4) added papers in areas the original search string was likely to undercount, with two refinements: prompt-injection and clinical work, originally excluded as topical, were re-admitted at rounds 2–3 because their *evaluation methodology*, not their content, is what the taxonomy classifies.

Each expansion round was triggered by a coverage gap observed after coding the prior cumulative set, was either exploratory (new topic area) or confirmatory (targeted retrieval), and introduces a directional bias toward its own target area, over-representing that area relative to the field’s true

Table 4. PRISMA 2020 corpus construction by round. All rounds applied the same inclusion criteria (§7.2). Round 0 used full database sweeps with dual-reviewer title/abstract screening (§7.3; $\kappa = 0.81$ on a 200-record stratified random subsample); Rounds 1–4 used narrower topical queries against the same six databases (so Identified = Screened, with retrieved records going directly to full-text review). *Round 0 arithmetic*: 412 advanced to full text, 356 excluded at full-text/venue review, 56 passed; 22 added via citation searching = 78 Added. Rounds 1–4’s procedural difference contributes to the selection bias named in §9.5. Columns: Identified after deduplication; Screened = full-text reviewed; Excluded = full-text + venue exclusions (reasons in repo); Added = retained; Cumulative = running corpus. Expansion rounds 1–4 contributed 73 / 151 papers (48.3%); a single broad primary search underestimates coverage for fast-moving topics.

Round	Target area(s)	Identified	Screened	Excluded	Added	Cumulative	Date
0	Primary PRISMA search + citation chase	2,847	412	356	78	78	2025-Q1–Q2
1	LLM-judge quality, contamination, prompt sensitivity	118	118	98	20	98	2025-Q3
2	Agentic, multilingual, prompt-injection, CoT faithfulness	74	74	63	11	109	2025-Q4
3	Foundation-model reports, RAG, safety, fairness, calibration, clinical, legal	212	212	177	35	144	2026-Q1
4	Long-context, tool-use, reward-model evaluation	46	46	39	7	151	2026-Q2
Total	–	3,297	862	733	151	151	2025-Q1–2026-Q2

prevalence (Electronic Supplement, Appendix E.6). Figure 3 shows the PRISMA 2020 flowchart aggregating all five rounds; Table 4 gives the per-round breakdown.

7.4 Data Extraction

For each paper we extracted: the prompting technique(s) evaluated; the seven dimensional codes D1–D7 (Section 4); the number of models compared; year, venue, and venue type; and a paper-category label (technique, benchmark, LLM-judge/metric, survey, or sensitivity/format/code-prompting) used to produce Table 8. The released <https://github.com/rohithreddybc/PromptEvalTaxonomy> repository carries the 151-row coding (D1–D7 only), its schema, the Appendix A checklist, and the analysis scripts.

7.5 Coding Reliability

Two independent coders extracted dimensional codes for the corpus. The primary coder coded all papers; the second coder independently re-coded a stratified random subsample of 38 papers (25% of the 151-paper corpus), stratified across paper categories, without access to the primary coder’s labels. Disagreements were resolved through discussion; cases that could not be resolved were referred to a third reviewer following PRISMA 2020 Item 8 [87].

Table 5 reports per-dimension agreement. Dimensions split into two regimes: D1–D5 reach $\kappa = 0.84–0.92$ (“almost perfect”, Landis–Koch threshold > 0.81); D6–D7 reach $\kappa = 0.76–0.78$ (“substantial”). The lower D6/D7 values reflect boundary ambiguity rather than factual disagreement (e.g., papers acknowledging sensitivity without quantifying it sit between D6=Not-reported and D6=Single-prompt). The heterogeneous profile is itself the finding: some properties are codable from descriptions, others require interpretive judgment that a reporting standard could reduce.

Ambiguity-resolution rules (full list in the released repository): D3 records the *primary* metric used for the headline claim; D4=Hybrid only if a spot-check conditions a reported decision (not when descriptive); D6=Sensitivity-bounded requires the headline pipeline to use the sensitivity

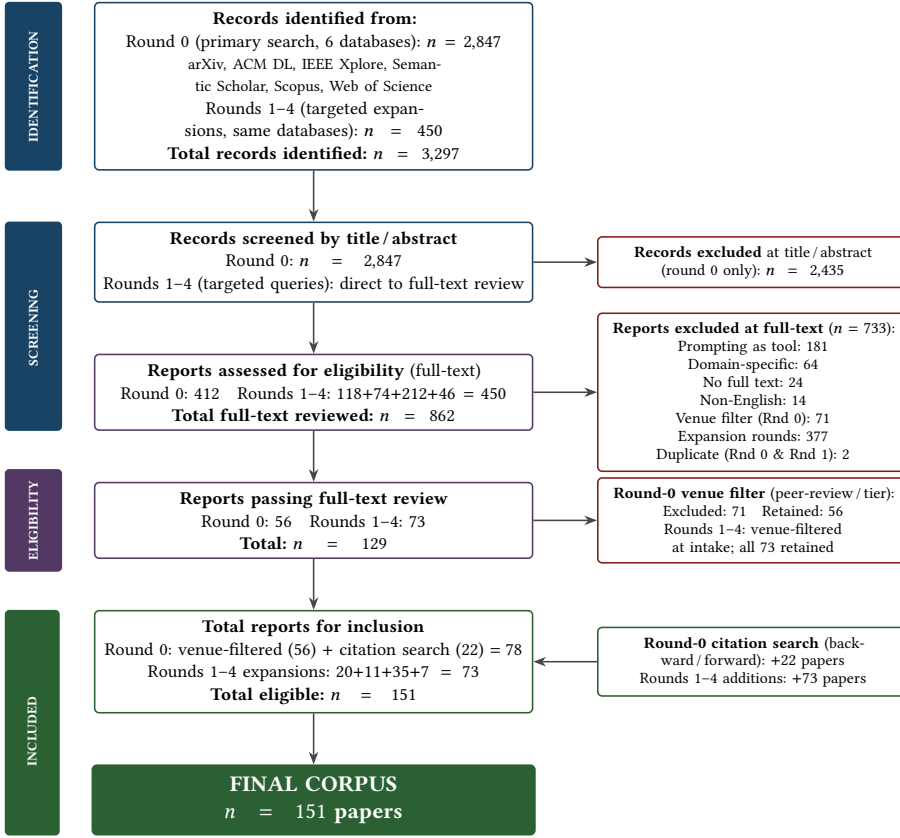


Fig. 3. PRISMA 2020 flowchart for all five search rounds combined (per-stage counts shown in the diagram). Of 3,297 records across six databases, 129 were included via search, augmented by 22 from round-0 citation searching for the 151-paper corpus. Per-round counts appear in Table 4; the diagram follows PRISMA 2020 reporting guidelines [87]. the high full-text exclusion (733) reflects papers that use prompting as a tool without evaluating it.

result, otherwise D6=Multi-prompt aggregate; D7=Contamination-checked requires an explicit verification step, not a generic statement of dataset differences.

7.6 Sensitivity Analysis

To test whether headline distributional claims are stable under coding uncertainty, we perturbed 10% and 20% of dimensional codes uniformly at random and re-derived the percentages (Table 6). All claims move by at most 3.4 percentage points under 20% perturbation; the largest shift (task-specific metric share, 3.4 pp) is borderline-stable relative to a 2-pp threshold and should be read with that caveat. Under 10% perturbation, all five claims shift by less than 2 pp.

7.7 Validity Threats

Three threats limit the scope of this review’s distributional claims: search-string coverage, our controlled vocabulary can miss work describing the same practice differently, and citation searching plus the four expansion rounds narrow but do not close that gap, so an absence here reflects our inclusion criteria, not the field [106, 109, 121]; the citation/venue quality filter, which favours

Table 5. Inter-rater agreement on dimensional codings, dual-coded subsample. Cohen’s κ per dimension, with interpretation following Landis & Koch. D1–D5 all reach “almost perfect” agreement ($\kappa \geq 0.84$, well above the Landis & Koch 0.81 threshold); D6–D7 reach “substantial” agreement ($\kappa = 0.76$ and 0.78), indicating robustness and reproducibility coding requires more interpretive judgment, as expected for properties not always explicitly reported in paper text.

Dimension	κ	Interpretation
D1 Evaluation method	0.92	Almost perfect
D2 Task scope	0.88	Almost perfect
D3 Metric type	0.91	Almost perfect
D4 Automation level	0.84	Almost perfect
D5 Evaluation scope	0.89	Almost perfect
D6 Robustness	0.76	Substantial
D7 Reproducibility	0.78	Substantial

Table 6. Stability of headline distributional claims under random coding perturbation. Reported values are from the 151-paper coding; shift bounds are the half-width of a 95% bootstrap interval derived by 10% and 20% random reassignment, 1,000 runs each. All headline distributional claims survive perturbation with shifts ≤ 3.4 pp under 20% random reassignment, suggesting the findings are stable under coding uncertainty and not artefacts of borderline cases.

Headline claim	Value	Shift (10/20%)
Benchmark-based evaluation share	72.2%	$\pm 1.5 / \pm 3.0$ pp
Fully automated evaluation share	81.5%	$\pm 1.2 / \pm 2.5$ pp
Task-specific metric share	67.5%	$\pm 1.8 / \pm 3.4$ pp
Clinical task representation	3.3%	$\pm 0.5 / \pm 1.1$ pp
Multilingual task representation	2.0%	$\pm 0.5 / \pm 1.0$ pp

work with time to accumulate citations and probably undercounts 2025–2026 papers in the fastest-moving subfields; and coding judgment at category boundaries, bounded by the inter-rater check (Section 7.5) and the perturbation analysis (Section 7.6), which together show how much the reported percentages could move if the borderline calls had gone the other way. Section 9.5 and Appendix E of the Electronic Supplement give the full scope of each threat.

8 Comparative Analysis

The corpus splits into two disjoint populations of 76 papers each: *PE-core* (technique, sensitivity/code prompting, PE-specific surveys) and *evaluation infrastructure* (benchmarks, datasets, LLM-judge frameworks, metric-quality studies). Aggregate distributions use the full corpus; the D7 table additionally reports the LLM-as-Judge subset.

Read every percentage below as a property of this corpus under the inclusion criteria of Section 7.2 (Section 9.5 details how far that generalizes); Archetype B concentrates specifically in LLM-judge frameworks. Per-dimension distributions reproduce by direct counts on the released coding; co-occurrence and dependency statistics require the analysis script (§10). The full 151-paper mapping, every paper cited, is Appendix C.

When the corpus is partitioned by paper category (a topical classification distinct from D1-atomic coding), the dimensional profile varies systematically (Table 8): technique papers (N=27), benchmark papers (N=44), and surveys (N=22) converge on Archetype A; the LLM-judge / metric-quality paper category (N=32, which includes the 30 papers with D1=LLM-as-Judge plus 2 metric-only studies) converges on B; sensitivity papers (N=27) show elevated D6 reporting. The per-category modal is informative *because of* the within-category variations; each variation is a candidate failure mode (Section 8.11) or methodological development (Section 9).

Table 7. Five most common evaluation-framework **configurations** in the corpus across the seven dimensions (raw counts; percentages over $N=151$). Inter-rater κ values are in Table 5. D7=Judge-version-pinned appears in only 5 corpus papers ($\approx 3.3\%$, Table 10), so LLM-as-Judge configurations below do not list JP in their modal D7. The configuration distribution shown here rolls up to the three-archetype distribution (Benchmark-Automation 66.2%, Judge-Mediated 17.2%, Expert-Anchored 9.3%, Mixed 7.3%) discussed in §8.10. The top five configurations account for 33.1% of the corpus; the choice between Benchmark-Automation and Judge-Mediated archetypes (95% CIs [58.9, 73.5]% and [11.3, 23.2]%) is the central design distinction.

Configuration (D1/D2/D3/D4/D5/D6/D7)	n	Share	Foregrounded tension
Benchmark / Multi-task / Task-spec / Auto / Cross-domain / Not reported / Not reported	13	8.6%	High reliability; no robustness or reproducibility reporting
Benchmark / Multi-task / Task-spec / Auto / Cross-model / Multi-prompt aggr. / Artifact-released	13	8.6%	High reliability with multi-prompt coverage
Benchmark / Multi-task / Task-spec / Auto / Cross-model / Single-prompt / Artifact-released	12	7.9%	High reliability, narrow construct validity
LLM-as-Judge / Multi-task / LLM-judge / Auto / Cross-model / Multi-prompt aggr. / Artifact-released	7	4.6%	Validity ambition with paraphrase coverage
Benchmark / Reasoning / Task-spec / Auto / Cross-model / Single-prompt / Artifact-released	5	3.3%	High reliability, narrow reasoning-task validity

Table 8. Category-aggregated mapping of the corpus to D1–D7. Modal configuration per category and within-category variations. All categories are modally automated in this corpus, and all but the survey category are modally cross-model; the primary differentiators are D3 (metric type) and D7 (artifact release), where LLM-judge papers have the highest artifact-release rate (86.7%).

Paper category	N	Modal D1/D3/D4	Modal D5	Notable within-category variation
Prompting technique (incl. APO)	27	Benchmark / Task-spec / Auto	Cross-model	A minority pair benchmark eval with hybrid automation (e.g. Ouyang et al. [86]).
Benchmark / dataset	44	Benchmark / Task-spec / Auto	Cross-model	Expansion-round benchmarks (IFEval [153], SWE-bench [55], WildBench [68]) are predominantly Cross-model; GLUE/MMLU/BIG-Bench form the Cross-dataset minority.
LLM-as-Judge / metric quality	32	LLM-as-Judge / LLM-judge / Auto or Hybrid	Cross-model	Expansion-round judge-bias papers (Koo et al. [59], Ye et al. [145]). D7=JP reported at materially higher rate.
PE / LLM-eval surveys	22	Benchmark / Task-spec / Auto	Cross-domain	Chang et al. [23] and Gehrmann et al. [42] anchor on human evaluation.
Sensitivity / format / code-prompting	27	Benchmark / Task-spec / Auto or Hybrid	Cross-model	Includes Mizrahi et al. [82] and clinical work (Sivarajkumar et al. [109]); D6 reaches its highest non-trivial rate here.

8.1 Distribution across Evaluation Method (Dimension 1)

Benchmark-based evaluation accounts for 109 papers (72.2%), the modal D1 in every paper-category subgroup of this corpus including foundation-model technical reports for GPT-4 [1], LLaMA [118,

[119], and GPT-4o [53]. Benchmarks are reproducible and cross-lab comparable but cannot capture output quality on open-ended tasks without a single correct answer.

LLM-as-Judge appears in 30 papers (19.9%), concentrated in open-ended generation and instruction-following (MT-Bench [150], AlpacaEval [34], G-Eval [72]). The cluster spans fine-tuned scalable judges [155], judge panels and meta-evaluation [24, 26, 124], two surveys [44, 63], bias quantification [115, 116, 133, 145], and hallucination detection [5]. Adoption has outpaced calibration research.

Human evaluation appears in 14 papers (9.3%; 10 atomic D1=Human eval plus 4 compound D1=LLM-as-Judge+Human dual-coded under both), ablation in 2 (1.3%); no primary user studies. Four papers [11, 21, 27, 143] are the compound cases, so subcategory counts sum to 155 while headline frequencies use the 151-paper denominator. The overall distribution favours scalable, reliable configurations over the construct-validity guarantees human evaluation provides (Section 9.1).

8.2 Distribution across Task Scope (Dimension 2)

Multi-task or cross-task evaluations account for 93 papers (61.6%): more than half the corpus does not assess techniques on a specific task type, instead aggregating across benchmarks. The pattern is partly an artefact of how survey and taxonomy papers classify, but also reflects a genuine tendency to report MMLU / BIG-Bench averages without per-task disaggregation. Reasoning is the largest task-specific subgroup at 14 papers (9.3%), reflecting chain-of-thought’s origin on arithmetic and symbolic benchmarks [135]; faithfulness-focused work in this subgroup [25, 60] treats the reasoning chain itself as the evaluation target. Generation appears in 10 papers (6.6%), code in 12 (7.9%), QA in 9 (6.0%), and agentic task evaluation in 4 (2.6%). These counts are atomic: one further paper is coded QA + Reasoning and is listed under neither, so the D2 subcategories partition the 151.

Clinical (5 papers, 3.3%) and multilingual (3 papers, 2.0%) are small populations; Table 6 reports perturbation-stability bounds for both (full per-paper breakdown in Appendix C). Under our inclusion criteria the corpus is strongly concentrated on multi-task benchmarks; the scarcity of clinical and multilingual papers reflects these criteria, not a field-level estimate [109, 121].

8.3 Distribution across Metric Type (Dimension 3)

Task-specific metrics dominate at 102 papers (67.5%) (pass@k, exact match, F1, accuracy); this tracks the D1 benchmark majority. LLM-judge metrics appear in 31 (20.5%); G-Eval [72] marked the shift to LLM-based scoring on open-ended outputs. Hybrid metrics in 16 (10.6%); model-based (BERTScore family) in 2 (1.3%), with [66] documenting the gradual shift to LLM-judge. Reference-based metrics (BLEU, ROUGE) appear as primary measures in zero corpus papers, a reversal of historical dominance following the lexical-overlap vs. human-judgment gap documented by [42].

8.4 Distribution across Automation Level (Dimension 4)

Fully automated evaluation is modal at 123 papers (81.5%); hybrid 19 (12.6%), manual 5 (3.3%), human-in-loop 4 (2.6%). The concentration reflects cost and throughput pressures and carries a D7 implication: pipelines depending on proprietary judges are stable within-run but not across-time, since the judge can change between API versions [39].

8.5 Distribution across Evaluation Scope (Dimension 5)

Cross-model is the most frequent scope at 96 papers (63.6%); cross-domain in 31 (20.5%, concentrated in surveys); single-model in 16 (10.6%); cross-dataset in 8 (5.3%). The two multilingual-topic papers [110, 122] are coded Cross-domain and Cross-model respectively under D5 (*evaluation scope*), despite their D2 being Multilingual. Single-prompt, single-model evaluations systematically overstate effects relative to multi-prompt baselines on the same model [82].

Table 9. Distribution of robustness reporting (D6), $N=151$. Inter-rater $\kappa = 0.76$; see Section 7.5. Single-prompt-only evaluation (41.1%) is most common; multi-prompt aggregate and sensitivity-bounded together account for 36.4%, elevated among D1=LLM-as-Judge papers (50.0%; see Section 8.8).

D6 subcategory	n	Share
Not reported	34	22.5%
Single-prompt	62	41.1%
Multi-prompt aggr.	46	30.5%
Sensitivity-bounded	9	6.0%
Total	151	100.0%

Table 10. Distribution of reproducibility properties (D7), $N=151$. Inter-rater $\kappa = 0.78$; see Section 7.5. Artifact release is a majority norm (70.9%) but contamination checking (4.6%) and judge-version pinning (3.3%) remain rare, a gap that will compound as LLM-as-Judge usage grows and model versions diverge from publication-time versions.

D7 property	All	LLM-judge ($N=30$)
Artifact-released	70.9%	86.7%
Contamination-checked	4.6%	3.3%
Judge-version-pinned	3.3%	16.7%
All three	0.0%	0.0%

8.6 Distribution across Robustness Reporting (Dimension 6)

Table 9 shows the D6 distribution ($\kappa = 0.76$; Section 7.5).

The sensitivity literature [13, 36, 82, 90, 97, 101, 105, 156] documents large paraphrase-induced variance and proposes formal sensitivity / consistency metrics, but this work has not propagated into the broader corpus as a routine reporting requirement. Hua et al. [52] offer a counterpoint: much of the reported sensitivity may be an artifact of rigid evaluation heuristics (log-likelihood scoring, exact-match) and reduces under LLM-as-Judge protocols, which itself reinforces the dependency of D6 conclusions on D1/D3 coding choices.

8.7 Distribution across Reproducibility (Dimension 7)

Table 10 shows the D7 distribution ($\kappa = 0.78$; Section 7.5).

The key asymmetry is between artifact-released share (70.9%) and contamination-checked share (4.6%): a paper that releases prompts and seeds without verifying test-set cleanness provides reproducibility artifacts while leaving construct unclear. Papers addressing this gap directly [31, 77, 100, 131, 140] are concentrated in the benchmark-paper category. Balloccu et al. [10] systematically document the scale of the problem: 4.7M samples from 263 benchmarks have leaked into GPT-3.5 / GPT-4 within their first year post-release, making the 4.6% contamination-checked share in our corpus more concerning, not less, when read alongside the documented leakage. Strong reproducibility requires both artifact release and contamination checking.

8.8 Cross-Dimensional Patterns

Read jointly rather than one at a time, the seven distributions show three patterns that no single-dimension table makes visible.

Reproducibility gaps cluster in benchmark papers, not LLM-judge papers. Of the 40 D7=Not-reported papers, 26 (65.0%) are D1=Benchmark and only 4 are D1=LLM-as-Judge; the LLM-judge cluster has higher artifact-release (86.7% vs. 70.9%) and concentrates judge-version pinning (16.7% vs. 3.3% overall), so the narrative that LLM-judge papers are the reproducibility laggard does not hold here. They also report multi-prompt sensitivity above average: 55 papers (36.4%) report multi-prompt aggregate or sensitivity-bounded robustness; 15/30 LLM-judge papers (50.0%) do so for the judge

itself. Yet 12/30 (40.0%) still use single-prompt judging, including AlpacaEval 2.0 [34], G-Eval [72], and FactScore [80].

Two-thirds of the corpus has neither robustness nor judge pinning, and FM3 is highly prevalent here. 94 papers (62.3%) carry both $D6 \in \{\text{Not-reported}, \text{Single-prompt}\}$ and $D7$ lacking Judge-version-pinned; any cross-paper score comparison there rests on two simultaneously unverified assumptions (paraphrase-stable technique, time-stable evaluator). 44 papers (29.1%) carry the FM3 fingerprint ($D5=\text{Cross-model}$ with $D6 \in \{\text{Single-prompt}, \text{Not-reported}\}$), including chain-of-thought [135]; cross-model claims under single-prompt support narrower conclusions than their phrasing suggests.

8.9 Co-occurrence Patterns Across Dimensions

The seven dimensions are conceptually independent but far from statistically independent. Table 11 reports pairwise co-occurrence for subcategory pairs the validity-reliability frame anticipates as linked: benchmark papers bundle metric and automation choices ($D1+D3 = D1+D4 = 66.2\%$), and LLM-as-Judge papers bundle with an LLM-judge metric (19.9%), confirming the judge-mediated archetype named later in this section.

Table 11. Selected pairwise co-occurrence rates across the corpus ($N=151$). Each cell is the share of papers exhibiting both properties. $D1$ – $D5$ values are exact; $D6$ and $D7$ values are subject to the inter-rater κ reported in Section 7.5. Seven pairs, chosen as those the validity-reliability frame predicts to co-occur, are shown; the full pairwise matrix is available in the machine-readable dataset.

Property A	Property B	Co-occurrence
$D1 = \text{Benchmark}$	$D3 = \text{Task-specific}$	66.2%
$D1 = \text{LLM-as-Judge}$	$D3 = \text{LLM-judge}$	19.9%
$D1 = \text{Benchmark}$	$D4 = \text{Automated}$	66.2%
$D5 = \text{Single-model}$	$D6 = \text{Single-prompt}$	8.6%
$D1 = \text{LLM-as-Judge}$	$D7 = \text{Judge-version-pinned}$	3.3%
$D2 = \text{Reasoning}$	$D3 = \text{Task-specific}$	9.3%
$D2 = \text{Generation}$	$D3 = \text{LLM-judge}$	4.6%

$D1/D3$ and $D5/D6$ entanglement are not coding artifacts but properties the dimensional structure makes visible: a benchmark paper typically adopts the benchmark’s metric and forgoes paraphrase-variance reporting. The validity-reliability frame organizes these clusters around coherent design-space regions with the same tradeoffs.

8.9.1 Dependency Strength: Cramér’s V across all dimension pairs. To test whether the apparent dimensional dependencies survive a formal independence test, we computed Cramér’s V for all 21 unordered dimension pairs using the primary atomic code per paper (first listed value when comma-separated). $V \in [0, 1]$ with conventional thresholds at 0.1 (weak), 0.3 (moderate), and 0.5 (strong) [29]. Table 12 reports the seven strongest pairs.

Three observations follow. Every listed pair clears the “moderate” threshold, so the entanglement pattern is not driven by one or two strong pairs; the seven dimensions form a connected dependency graph. The strongest pair ($D1$ – $D3$) reflects the validity-reliability frame’s prediction that judgment locus and metric type are linked: benchmarks commit to reference-based metrics, LLM-judges to LLM-judge metrics. $D5$ – $D6$, $D5$ – $D7$, and $D6$ – $D7$ form a connected cluster, consistent with reproducibility and robustness reporting sharing an infrastructural cause: papers pinning their

Table 12. Cramér’s V dependency strength for the seven most strongly associated dimension pairs (of 21 total pairs), with underlying χ^2 test statistic and degrees of freedom. All seven clear the 0.3 “moderate-association” threshold; the top four clear 0.49 (close to or above “strong”). The full pairwise table is in the released machine-readable dataset. *Note on df*: contingency tables are computed under the primary-atomic-code rule (§8.9.1); compound D7 codes (JP+AR, CC+AR) are collapsed to their primary property (JP, CC), giving D7 four categories (AR, NR, JP, CC) and $df = 9$ for D5–D7 and D6–D7. Raw values reproduce from `empirical_analysis.ps1` on `corpus_d1d7.csv`. The strongest dependencies (D1–D3 = 0.701; D1–D4 = 0.589) confirm that evaluation-method choice is strongly associated with metric and automation choices in a statistically detectable pattern; the dimensions are non-redundant but not independent, which is consistent with what the archetype analysis identifies.

Dimension pair	Cramér’s V	χ^2	df
D1 (Method) – D3 (Metric)	0.701	222.9	9
D1 (Method) – D4 (Automation)	0.589	157.3	9
D5 (Scope) – D6 (Robustness)	0.497	112.0	9
D6 (Robustness) – D7 (Reproducibility)	0.491	109.2	9
D5 (Scope) – D7 (Reproducibility)	0.490	108.6	9
D3 (Metric) – D4 (Automation)	0.397	71.5	9
D2 (Task) – D3 (Metric)	0.383	66.6	21

judge version tend to release artifacts and report multi-prompt aggregates. The weakest pair (D4–D6, $V = 0.189$) confirms that automation level says little about paraphrase variance reporting. The full matrix is consistent with the archetype account: the seven dimensions are non-redundant descriptors that are nevertheless correlated in the literature, which is precisely what produces recurring archetypes.

8.10 Three Recurring Evaluation Archetypes

The co-occurrence pattern resolves into three recurring archetypes that, taken together, cover 92.7% of the corpus. Each is named for a region of the validity-reliability space.

The assignment rule. Archetypes are assigned from **D1 and D4 only**, in a fixed precedence order (C, then B, then A), so that each paper receives at most one label: C if D1 includes Human evaluation; otherwise B if D1=LLM-as-Judge; otherwise A if D1=Benchmark and D4=Automated; otherwise *Mixed*. The 7.3% (11 papers) matching no clause are the Mixed residual, which is why the three shares sum to 92.7% rather than 100%. Applying this rule to `corpus_d1d7.csv` reproduces every share below exactly.

On the 140 papers that receive an archetype, D4 never changes a label: every D1=Benchmark paper is A, every D1=LLM-as-Judge paper is B, and every paper whose D1 includes human evaluation is C. D4’s only effect is to route 9 Benchmark-with-Hybrid-automation papers into Mixed, and Mixed is excluded from the tests that follow. The archetypes are therefore best read as a three-way collapse of D1, not as a genuinely two-dimensional construct.

The “modal profile” lines below are *descriptive*, not part of the rule: they say what a typical paper in each archetype looks like along the other dimensions, and individual papers deviate. Per-archetype 95% CIs are percentile-method intervals over 10,000 bootstrap resamples, RNG seed 42 (`empirical_analysis.ps1`). That Mixed and FM1 both cover 11 papers is a coincidence of denominators; the two sets share only 2 papers.

Archetype A: Benchmark-Automation. Assigned by: D1=Benchmark and D4=Automated. Modal profile: D3=Task-specific, D6=Single-prompt or Not-reported, D7=Artifact-released. Share:

66.2% (100 papers; 95% CI [58.9%, 73.5%]; modal for technique, benchmark, and survey categories). Region: high reliability, narrow construct validity, reference-anchored. Representative: chain-of-thought [132, 135] and reasoning benchmarks [49, 111]. Strength: reproducible, cross-lab comparable, well-matched to closed-form reasoning. Weakness: under-specifies the construct on open-ended generation; sensitive to contamination; rarely reports paraphrase variance.

Archetype B: Judge-Mediated. Assigned by: D1=LLM-as-Judge. Modal profile: D3=LLM-judge, D4=Automated or Hybrid, D5=Cross-model. Share: 17.2% (26 papers; 95% CI [11.3%, 23.2%]; concentrated in LLM-judge and metric-quality categories). Region: judgment-anchored, broader construct ambition than A, but reliability mediated by the underlying judge. Representative: MT-Bench [150], AlpacaEval [34], LLM-judge analyses [21, 39, 44]. Strength: handles open-ended outputs where A cannot. Weakness: judge bias and version drift are not yet managed by a shared calibration protocol.

Archetype C: Expert-Anchored. Assigned by: D1 includes Human evaluation (sometimes plus LLM-judge). Modal profile: D3=Hybrid, with a substantial LLM-judge minority; D4=Human-in-loop or Hybrid. Share: 9.3% (14 papers; 95% CI [4.6%, 14.6%]; the wider relative interval reflects the small count). Region: judgment-anchored, highest construct ambition, lowest within-run reliability. Representative: InstructGPT/RLHF [86], human-vs-LLM calibration [11, 27, 143]. Strength: takes construct validity seriously where reference-anchored evaluation is structurally inadequate. Weakness: scales poorly; inter-rater agreement is rarely strong.

8.10.1 Cluster Validation: are the archetypes real? Four checks on the one-hot-encoded D1–D7 codes ask whether the archetypes reflect structure or imposed labels; Appendix H of the Electronic Supplement gives the full method and results. In summary: the rule-based labelling on the 140 archetype-fitting papers has silhouette **0.414** against -0.037 for random three-label assignments; unsupervised k -means at $k = 3$ agrees with it at ARI = 0.50; across five algorithms ARI ranges 0.49–0.90; and bootstrap co-cluster probability is 0.73/0.89/0.65 within archetypes against 0.09 between. Under k -means, silhouette is marginally higher at $k = 4$ (0.358) than $k = 3$ (0.348), so we take $k = 3$ on parsimony and interpretability, not because it maximises silhouette.

These checks are weaker than they look. All four cluster the full D1–D7 encoding, which *contains* D1, the dimension the rule reads, so the recovery is largely the algorithm reading back its own input: D1 alone reproduces the archetypes at ARI ≈ 0.97 , at or above what the five algorithms achieve on the full encoding, while the five dimensions the rule does *not* read (D2, D3, D5–D7) recover it at ARI ≈ 0.01 – 0.02 , i.e. not at all. The archetypes are a compact summary of D1, not a latent structure the other dimensions independently corroborate.

8.10.2 Predictive Validation: does the archetype carry information beyond its construction? We tested whether archetype labels are associated with D6 and D7, neither of which the assignment rule reads, on the 140-paper archetype-fitting subset (11 mixed excluded), using primary-atomic D7 so both tests use 6 df. **The test does not come out in our favour, and we report it as it stands.**

$\chi^2 = 11.83$, $V = 0.21$ for archetype $\times$ D6; $\chi^2 = 30.39$, $V = 0.33$ for D7. Several cells violate Cochran’s rule (expected < 5 in 4/12 D6 cells, 7/12 D7 cells), so we add Monte Carlo p -values from 10,000 permutations under fixed marginals (RNG seed 42, reproducible via `recompute_all.py`): $p_{MC} = 0.066$ (D6), $p_{MC} = 0.0005$ (D7). After Bonferroni ($m=2$), D6 is not significant ($p_{adj} = 0.132$).

The apparently robust D7 result ($p_{adj} < 0.002$) does not survive scrutiny. Its strongest cell is judge-version pinning: 0% in Archetype A (0/100) vs. 19.2% in B (5/26). But D7=JP records the version of an *LLM-judge* component (Definition 7), and Archetype A is assigned on D1=Benchmark, i.e. scoring with no judge (Definition 4). A paper with no judge cannot pin a judge version, so the 0/100 is closer to a logical necessity than a discovery: D6 and D7 are not *inputs* to the rule, but the *support* of D7 is conditioned on D1. With the JP level removed, on the 135 papers whose D7 levels are reachable from every archetype, $\chi^2 = 7.52$, $df = 4$, $V = 0.17$, $p = 0.11$: not significant.

We therefore withdraw the predictive-validity claim. The dimension genuinely held out (D6) is null after correction; the one that appears to work (D7) does so through a structural dependency on the assignment input. What remains is descriptive: reproducibility reporting differs across archetypes (D7=Not-reported: 57.1% of C, 25.0% of A, 15.4% of B).

8.11 Failure-Mode Catalog

Beyond distributional patterns, the seven-dimensional coding identifies cases where a paper’s evaluation configuration does not fully support its contribution claim. We name three failure modes (FM1: Construct Mismatch; FM2: Unpinned Judge; FM3: Single-Phrasing Generalization), give the dimensional fingerprint of each, and identify named corpus papers carrying that fingerprint. The intent is diagnostic, not accusatory: each named paper makes a defensible contribution within its own scope, and naming the dimensional gap is not the same as discounting the contribution.

Table 13 gives the fingerprint, prevalence, mitigation, and one exemplar per failure mode.

FM1: Construct Mismatch. Beyond Sivarajkumar [109], Nori et al. [85] (clinical) carries the same fingerprint and Hou et al. [50] (code) exhibits a closely related pattern outside the FM1 D2 subset. The benchmark metric supports answer-key agreement, but the deployment-readiness claim requires Archetype-B/C evidence or a calibrated reference-free metric that these papers do not provide; the lexical-overlap vs. human-judgment gap is documented in [42, 72].

FM2: Unpinned Judge. Besides Thakur et al. [116], Bavaresco et al. [11] and Chen et al. [24] show the same fingerprint. Within-run reproducibility is intact but across-time is not, because the proprietary judge can be silently revised; judge-drift quantification [13, 39] confirms the concern, and only the 5 corpus papers coded D7=Judge-version-pinned (3.3% overall, 16.7% of LLM-judge) document the snapshot.

FM3: Single-Phrasing Generalization. Wei et al. [135] vary exemplar sets and annotators but not the instruction phrasing, so the cross-model claim rests on one phrasing per model; zero-shot CoT [58], by contrast, tests 16 templates and is coded Multi-prompt aggregate. Mirzadeh et al. [81] report accuracy declines when GSM8K numerical values are changed and drops of up to 65% when a seemingly-relevant but inert clause is added (their GSM-NoOp condition); Mizrahi et al. [82] and Salinas & Morstatter [101] show single-prompt overstatement under paraphrase; Vaugrante et al. [123] report that CoT, EmotionPrompting, and ExpertPrompting gains lose statistical significance once replicated across six models on multiple reasoning benchmarks. FM3 is not concentrated in any one archetype: it affects 30.0% of Archetype A (30/100) against a corpus rate of 29.1%, so it is a corpus-wide reporting habit rather than a property of benchmark-automation work specifically.

Aggregate prevalence. 68 of 151 papers (45.0%) exhibit at least one of FM1–FM3; overlap is modest ($FM1 \cap FM3 = 3$; $FM2 \cap FM3 = 9$; no paper exhibits all three or both FM1 and FM2). These counts are reproducible from corpus_d1d7.csv by applying the fingerprints above; any reader can compile a similar list under a different fingerprint.

Table 13 consolidates FM1–FM3 as a single-page reference that subsequent work can cite when reporting how a study addressed a given failure mode (e.g., “we mitigate FM2 by recording the judge snapshot in Section X”).

Hidden assumptions the dimensional view exposes. Each failure mode names an unstated assumption supporting the headline claim, and coding makes five of them visible: *construct equivalence* (D2+D3 puts a task-specific metric on a generation task), *judge stability* (D1=LLM-as-Judge but D7 does not pin the judge), *paraphrase invariance* (D6=Single-prompt under cross-model scope), *contamination cleanliness* (D7 lacks contamination-checked), and *category representativeness* (D2=Multi-task with no disaggregation). None are novel in isolation [39, 82, 84, 105]; the taxonomy locates them in specific papers and quantifies how often each goes unaddressed.

Table 13. Failure-mode quick-reference. Fingerprint = dimensional pattern; prevalence and exemplar reproducible from corpus_d1d7.csv. *Citation guidance:* a paper addressing FMx can cite this table for fingerprint and mitigation. Each FM has a one-step reporting-change mitigation; 45.0% of the corpus exhibits at least one FM.

FM	Name	Fingerprint	Prevalence	Mitigation	Exemplar
FM1	Construct Mis-match	D1=Benchmark, D3∈{Task-spec., Model-based} on D2∈{Gen, Clin, Agentic, Multilingual}	11 / 151 (7.3%)	Calibrated reference-free metric (e.g., FActScore [80]) or human anchor	Sivarajkumar [109]
FM2	Unpinned Judge	D1≥LLM-as-Judge; D7 lacks JP	25 / 30 LLM-judge = 83.3% (16.6% corpus)	Record judge model + snapshot date; pin check-point when possible	Thakur et al. [116]
FM3	Single-Phrasing Generalization	D5=Cross-model D6∈{Single-prompt, NR}	+ 44 / 151 (29.1%)	Variance over ≥ 5 phrases [82, 156]	Wei [135]

9 Discussion and Future Directions

The distributions (Section 8) motivate the two tensions of Section 4.1, with a within-run vs. across-time refinement (Definition 3); the archetype analysis adds no independent support (§8.10.1).

9.1 Tension 1: Scalability vs. Construct Validity

Archetype A maximizes within-run reliability and throughput by anchoring to a fixed gold reference; Archetype C maximizes construct validity by anchoring to human judgment, at the cost of replication speed and cross-paper comparability. The corpus is heavily weighted toward A (66.2% vs. 9.3%). Both archetypes are internally rigorous; the tradeoff between them is *construct fidelity at deployment* against *measurement consistency across runs*. On tasks an exact-match score approximates well (arithmetic, factual retrieval, code execution), Archetype A loses little validity for high reliability; on tasks it does not (open-ended generation, clinical decision support, multilingual fluency), the same reliability comes with a score that under-determines what it claims to measure, so deployment claims there should not rest on Archetype-A evaluation alone.

9.2 Tension 2: Reference-Anchored vs. Judgment-Anchored Evaluation

The second tension marks the boundary between Archetypes A and B: without a gold reference, scoring delegates to a judgment (human, expert panel, or LLM), so reference-anchored evaluation fails on under-determined references and contamination while judgment-anchored evaluation fails on judge bias, version drift, and missing calibration. The 30 LLM-judge papers use at least six distinct judge configurations (model, prompt template, rubric, tie-handling) with few reporting calibration against a target judgment, so their scores are reproducible within a paper but not comparable across papers. Calderon et al.’s judge–annotator substitutability framework [21] and Gehrmann et al.’s broader NLG reporting norms [42] remain unadopted at scale within PE evaluation.

9.3 Refining Tension 1: Within-Run vs. Across-Time Reliability

A pipeline can be reliable within-run yet fragile across-time (API revisions drift; test data leaks into pretraining). D7 captures this: artifact release is common, but contamination-checking and judge-version pinning are not (Section 8.7). Dynamic contamination-resistant benchmarks [68] and judge-sensitivity quantification [13, 39] address parts of this; broader uptake remains open.

9.4 Open Problems by Tension

Each tension implies an open problem; we label them OP1–OP3 so subsequent work can cite the specific gap it addresses. **OP1** (scalability vs. validity): *calibrated reference-free metrics*, validated against an expert standard; FActScore [80] and the alternative annotator test [21] are building blocks, but a systematic domain-specific PE evaluation literature does not yet exist. **OP2** (reference vs. judgment): a *shared LLM-judge calibration standard* with minimum disclosure (judge model, version, prompt template, calibration outcome). **OP3** (within-run vs. across-time): *infrastructure-stable pipelines* via dynamic contamination-resistant benchmarks, locally pinned judge models, or at minimum snapshot-date reporting for proprietary components.

9.5 Limitations of This Survey

The findings are bounded by five typed limitations, each scoped in detail in Appendix E of the Electronic Supplement: (1) *search-string coverage* (controlled-vocabulary limits plus expansion-round selection bias toward targeted areas); (2) *recency bias* against 2025–2026 work without venue or citations (mitigated by extended cutoffs but not eliminated); (3) *coding-judgment ambiguity* on D4/D6/D7 boundary cases (quantified by per-dimension κ); (4) *corpus scope*, which excludes domain-specific applied papers and non-English work (absence in corpus is not absence in field); and (5) *expansion-round bias* toward LLM-judge, agentic, foundation-model, and long-context targets. Distributional findings are descriptive of this corpus under stated criteria, not field-level estimates; the sensitivity analysis (Section 7.6) quantifies how much each limitation propagates to headline claims.

9.6 A Future Research Agenda

Seven research directions follow from the tensions and open problems identified above: (1) *domain-specific frameworks* with expert-calibrated metrics for clinical, legal, and multilingual PE; (2) an *LLM-judge calibration standard* with common disclosure; (3) *scope as minimum bar* (multi-model, multi-benchmark, multi-paraphrase by default); (4) *robustness reporting* of sensitivity bounds alongside point estimates, following statistical practice for evaluation uncertainty more broadly [79]; (5) *agentic/multimodal conventions* for D1/D3/D6; (6) *reporting templates*, e.g., a venue-endorsed checklist of model, seed, prompt, and judge; and (7) *safety and bias integration* as first-class dimensions. The mapping of each direction to its primary gap, affected dimensions, and key references is given in Appendix G of the Electronic Supplement.

Reasoning-first models (o1, o3, DeepSeek-R1) deserve more space: they challenge D1 (cannot attribute performance to prompt design versus autonomous internal CoT), D3 (an open measurement problem in trace faithfulness), and D6 (a new variance source from internal CoT sampling invisible to single-prompt reporting). Whether these three need new subcategories is a priority for the next survey cycle.

9.7 Using This Taxonomy in Citing Work

Four non-prescriptive applications follow for citing work: embedding the seven-tuple in a single methodology line (e.g. “D1 = Benchmark, D2 = Reasoning, D3 = Task-specific, D4 = Automated, D5 = Cross-model, D6 = Multi-prompt-aggregate, D7 = Artifact-released”); using the machine-readable corpus dataset (Data Availability) as a comparison reference rather than re-coding papers; using the Electronic Supplement (Appendix A) checklist as a preregistration template; and citing Table 13 when an evaluation design addresses FM1, FM2, or FM3. Canonical anchors: D1–D7 (Sections 4.3–4.9), archetypes (Section 8.10), FM1–FM3 (Table 13), OP1–OP3 (Section 9.4), and the checklist (Table 3, Appendix A).

10 Conclusion

This survey treats prompt engineering evaluation as a measurement problem rather than a procedural choice: two interpretive tensions from measurement theory, construct validity against measurement reliability, and reference-anchored against judgment-anchored evaluation, organize the space of evaluation choices a study makes, and a small set of dimensions gives those tensions operational form. Applied to a systematic review of the literature, the taxonomy shows that evaluation practice clusters into a handful of recurring configurations rather than spreading evenly across the space it defines, that the empirical structure separating those configurations traces back to a single dimension rather than to the taxonomy as a whole, and that several recurring combinations of evaluation choices leave a paper unable to fully support the claim it makes with them. The taxonomy does not settle which evaluation method is optimal for a given task, nor does it supply domain expertise for specialized protocols. What it establishes is a shared vocabulary in which the tradeoffs prompt engineering evaluation makes, usually silently, become nameable, comparable across studies, and open to methodological argument rather than left to convention.

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Data Availability

The companion artifact repository is publicly available at <https://github.com/rohithreddybc/PromptEvalTaxonomy> under a CC-BY 4.0 license. The repository provides the 151-paper $\times$ D1–D7 coding dataset (`corpus_d1d7.csv`) with its column schema (`dataset_readme_new.md`), the Electronic Supplement (`supplementary.pdf`, including the Appendix A evaluation-design checklist), the PowerShell script reproducing the empirical analyses in §8.9.1/§8.10.1 (`empirical_analysis.ps1`), the Python scikit-learn cluster validation script (`cluster_validation.py`), and the PRISMA flow record (`prisma_flowchart.png` with generator `make_prisma.py`).

Review-time artifact access. An anonymized snapshot of the coding spreadsheet, the empirical-analysis script, and the PRISMA flow record ships with the submission package and at the repository URL above, so reviewers can audit individual codings, regenerate the silhouette and Cramér’s V values of §8, and verify the per-round PRISMA counts during review.

At acceptance the repository will be deposited to Zenodo with a persistent, version-aligned DOI.

Claim-to-artifact mapping. The repository README maps each empirical claim to the artifact reproducing it: per-dimension distributions and archetype shares from `corpus_d1d7.csv`; failure-mode prevalence from the same file under the Section 8.11 fingerprints; Cramér’s V and χ^2 from the empirical-analysis script; and silhouette, ARI, and bootstrap stability from `cluster_validation.py` (seed = 42). An integrity-audit script re-verifies in one command that the coding matches the released CSV and that every cited key resolves.

Author Contributions, Funding, and Conflict of Interest

This work was conducted independently by the authors. No external funding supported this work. The authors declare no competing financial or non-financial interests. Author contributions follow the CRediT taxonomy: R.R.B. conceived the validity/reliability framing, led methodology and primary coding, and drafted the manuscript; P.J. and T.M. contributed inter-rater coding, ambiguity resolution, and review; E.R. provided conceptual guidance and critical revision; W.Z. provided research supervision and senior review.

For full transparency with respect to the ACM Conflict of Interest Policy, the authors disclose that co-author W. Zhang serves on the Editorial Board of ACM Computing Surveys as an Associate Editor at the time of submission. W. Zhang will not participate in any editorial decisions, reviewer selection, or handling of this manuscript. The Editor-in-Chief is asked to assign a handling editor with no collaboration history with any of the authors. This disclosure is provided in accordance with the ACM Peer Review Policy.

Ethical Considerations

This work is a meta-level systematic review conducted entirely over publicly available published and preprint papers. No human subjects research was conducted, and no personally identifiable information was collected or processed. All cited work is attributed accurately to the original authors with complete bibliographic records. The survey does not involve experiments on human participants, animal subjects, or sensitive data; accordingly, no IRB review was required. Potential dual-use concerns are mitigated by the descriptive and analytical nature of the survey, which characterizes existing evaluation methodology rather than introducing new attack capabilities.

Use of AI Assistants

Generative AI assistants (large language models accessed via commercial APIs) were used during manuscript preparation for routine editing tasks: rephrasing for clarity, table-formatting code generation, and bibliographic-entry formatting. All scientific content, taxonomy definitions, methodological choices, paper coding, and analytical conclusions are the authors' own work. No AI-generated text was incorporated without authorial review and revision. Consistent with ACM policy, AI assistants are not credited as authors.

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